Reg. No. | | | | | | | | | | |

[21MADIP02/18MADIP02]

FOURTH SEMESTER B.Tech EXAMINATION, JUNE-2023

DIPLOMA MATHEMATICS-II

Time:3 Hours

Maximum Marks:70

SL. NO.	QUESTIONS	A	B	C	D	ANSWER
1.	$\int_2^4\int_0^1 dxdy =$	2	4	8	None of these	
2.	$\int_2^3\int_{-1}^1 xdxdy =$	2	5	6	0	
3.	$\int_2^3\int_1^2 ydxdy =$	$\frac{7}{2}$	$\frac{3}{2}$	$\frac{5}{2}$	$\frac{9}{2}$	
4.	$\int_{-2}^3\int_0^1 xydxdy =$	$\frac{9}{2}$	$\frac{5}{4}$	$\frac{7}{2}$	None of these	
5.	$\int_4^5\int_2^3 x^2ydxdy =$	$\frac{57}{2}$	$\frac{59}{2}$	$\frac{9}{2}$	0	
6.	$\int_1^{-1}\int_{-2}^2 xy^2dxdy =$	0	3	4	14	
7.	$\int_3^{-4}\int_1^2 (x+y)dxdy =$	14	0	4	−14	
8.	$\int_1^{-1}\int_0^{\frac{\pi}{2}} \cos xdxdy =$	0	3	−2	1	
9.	$\int_1^4\int_0^{\infty} e^{-x}dxdy =$	1	2	4	3	
10.	$\int_{-2}^2\int_0^{-\infty} ye^xdxdy =$	2	0	3	6	
11.	$\int_{-1}^2\int_0^{\frac{\pi}{2}} \sin y\, dydx =$	3	4	2	−2	
12.	$\int_0^2\int_0^{\pi} x\sin y\, dydx =$	−2	2	4	3	
13.	$\int_2^3\int_0^{\frac{\pi}{2}} \sin^2 x\, dxdy =$	$\frac{\pi}{2}$	$\frac{\pi}{4}$	π	$\frac{\pi}{3}$	
14.	$\int_0^1\int_0^{\frac{\pi}{4}} \sec^2 x\, dxdy =$	8	1	6	None of these	
15.	The Laplace Transform of $f(t)$ is	$\int_0^{\infty} e^{-st} f(t)dt$	$\int_0^{\infty} e^{-t} f(t)dt$	$\int_0^{2a} e^{-st} f(t)dt$	None of these	
16.	The value of $L[1]$ is	$\frac{4}{s}$	$\frac{3}{s}$	$\frac{2}{s}$	$\frac{1}{s}$	

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17.	$L(e^{at})$ is equal to	$\frac{1}{s-a}$	$\frac{1}{s+a}$	$\frac{2}{s-a}$	$\frac{2}{s+a}$	
18.	$L(e^{-at})$ is equal to	$\frac{1}{s-a}$	$\frac{1}{s+a}$	$\frac{2}{s-a}$	$\frac{2}{s+a}$	
19.	$L(\sin at)$ is equal to	$\frac{a}{s^2-a^2}$	$\frac{a}{s^2+a^2}$	$\frac{s}{s^2-a^2}$	$\frac{s}{s^2+a^2}$	
20.	$L(\cos at)$ is equal to	$\frac{a}{s^2-a^2}$	$\frac{a}{s^2+a^2}$	$\frac{s}{s^2+a^2}$	$\frac{s}{s^2-a^2}$	
21.	$L(\sinh at)$ is	$\frac{1}{s^2+a^2}$	$\frac{a}{s^2-a^2}$	$\frac{s}{s^2+a^2}$	None of these	
22.	$L(\cosh at)$ is	$\frac{1}{s^2+a^2}$	$\frac{1}{s^2-a^2}$	$\frac{s}{s^2+a^2}$	$\frac{s}{s^2-a^2}$	
23.	$L(t^n)=$, Where n is non integer	$\frac{\Gamma(n)}{s^{n+1}}$	$\frac{\Gamma(n-1)}{s^{n+1}}$	$\frac{\Gamma(n+1)}{s^{n+1}}$	None of these	
24.	$L(t^n)=$, Where n is integer	$\frac{\Gamma(n-1)}{s^{n+1}}$	$\frac{n!}{s^{n+1}}$	$\frac{\Gamma(n)}{s^{n+1}}$	None of these	
25.	$L(e^{3t})$ is equal to	$\frac{1}{s-3}$	$\frac{1}{s+3}$	$\frac{2}{s-3}$	$\frac{1}{s-a}$	
26.	$L(e^{-4t})$ is equal to	$\frac{4}{s+4}$	$\frac{2}{s-4}$	$\frac{1}{s-a}$	$\frac{1}{s+4}$	
27.	Laplace transform of $\sin 2t$ is	$\frac{2}{s^2-4}$	$\frac{2}{s^2+4}$	$\frac{2}{s^2+2}$	$\frac{2}{s^2-2}$	
28.	Laplace transform of $\cos 5t$ is	$\frac{5}{s^2+25}$	$\frac{5}{s^2-25}$	$\frac{s}{s^2+25}$	$\frac{s}{s^2-25}$	
29.	Mathematically, the functions in Green's theorem will be	Continuous derivatives	Discrete derivatives	Continuous partial derivatives	Discrete partial derivatives	
30.	Find the value of Green's theorem for $F = x^2$ and $G = y^2$ is	0	1	2	3	
31.	Which of the following is not an application of Green's theorem?	Solving two-dimensional flow integrals	Area surveying	Volume of plane figures	Centroid of plane figures	
32.	The path traversal in calculating the Green's theorem is	Clockwise	Anticlockwise	Inwards	Outwards	
33.	Calculate the Green's value for the functions $F = y^2$ and $G = x^2$ for the region $x = 1$ and $y = 2$ from origin.	0	2	-2	1	
34.	If two functions A and B are discrete, their Green's value for a region of circle of radius a in the positive quadrant is	∞	$-\infty$	0	Does not exist	
35.	Applications of Green's theorem are meant to be in	One dimensional	Two dimensional	Three dimensional	Four dimensional	
36.	The Green's theorem can be related to which of the following theorems mathematically?	Gauss divergence theorem	Stoke's theorem	Euler's theorem	Leibnitz's theorem	
37.	Find the area of a right angled triangle with sides of 90 degree unit and the functions described by $L = cosy$ and $M = sinx$.	0	45	90	180	
38.	Gauss theorem uses which of the following operations?	Gradient	Curl	Divergence	Laplacian	
39.	Evaluate the surface integral $\iint(3xi + 2yj)dS$, where S is the sphere given by $x^2 + y^2 + z^2 = 9$	120π	180π	240π	300π	

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40.	The Gauss divergence theorem converts	line to surface integral	line to volume integral	surface to line integral	surface to volume integral	
41.	The divergence theorem for a surface consisting of a sphere is computed in which coordinate system?	Cartesian	Cylindrical	Spherical	Depends on the function	
42.	Find the Gauss value for a position vector in Cartesian system from the origin to one unit in three dimensions.	0	3	-3	1	
43.	$\frac{d}{dz}[f \pm g] =$	$\frac{df}{dz} \pm \frac{dg}{dz}$	$\frac{df}{dz} \pm \frac{df}{dz}$	$\frac{df}{df} \pm \frac{dg}{df}$	$\frac{df}{dg} \pm \frac{dg}{dg}$	
44.	$\frac{d}{dz}[fg] =$	$f\frac{dg}{dz} \times g\frac{dg}{dz}$	$f\frac{dg}{dz} + g\frac{df}{dz}$	$f \pm g\frac{dg}{dz}$	$\frac{dg}{dz} \pm \frac{dg}{dz}$	
45.	$\frac{d}{dz}[\frac{f}{g}] =$	$\frac{g - f\frac{dg}{dz}}{g^2}$	$\frac{g\frac{df}{dz} - f\frac{dg}{dz}}{}$	$\frac{gf\frac{dg}{dz}}{g^2}$	$\frac{g\frac{df}{dz} - f\frac{dg}{dz}}{g^2}$	
46.	$\frac{d}{dz}[f(z)]^n =$	$n[f(z)]^{n-1}$	$[f(z)]^{n-1}\frac{df}{dz}$	$n[f(z)]^{n-1}\frac{df}{dz}$	$n[f(z)]\frac{df}{dz}$	
47.	$\frac{d}{dz}[z]^n =$	nz^{n-1}	nz	z^{n-1}	$7z^{n-1}$	
48.	$f(z)$ is differentiable, $f'(z) =$	$\frac{f(z+\Delta z) - f(z)}{\Delta z}$	$\lim_{\Delta \to 0}\frac{f(z+\Delta z) - f(z)}{\Delta z}$	$\lim_{\Delta \to 0} z$	$\lim_{\Delta \to 0}\frac{f(z+\Delta z)}{\Delta z}$	
49.	A function is analytic in a domain if	It is analytic at not every point of the domain	It is analytic at a point of the domain	It is analytic at every point of the domain	None of these	
50.	An analytic function is also known as	Differentiable	continuity	Not regular	regular	
51.	$f(z) = u + iv$ then $\lvert f(z)\rvert =$	$\sqrt{u+v}$	$\sqrt{u^2+v^2}$	uv	$\sqrt{u^2-v^2}$	
52.	$f(z) = u + iv$ then $f'(z) =$	$u_x + iv_x$	$\sqrt{u^2-v^2}$	$u_x v_x$	$\sqrt{u^2v^2}$	
53.	Cauchy's Reimann equation in cartesian form is	$u_x = v_y \,\&\, v_x = -u_y$	$v_x = -u_y$	$u_x = v_y$	$u_x v_x$	
54.	$\lvert 4+i3\rvert =$	5	7	3	2	
55.	$\lvert 4+i\rvert =$	$\sqrt{4}$	$\sqrt{17}$	$\sqrt{8}$	$\sqrt{15}$	
56.	$\lvert 5+i4\rvert =$	4	9	6	None of these	
57.	Which of the following matrices is in row reduced echelon form?	$\begin{bmatrix} 0 & 1 & 9 \\ 0 & 0 & 1 \\ 2 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 3 & 0 & 7 \\ 0 & 1 & 5 & 6 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 3 & 6 & 5 \\ 2 & 6 & 12 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix}$	None of these	
58.	Let $A = \begin{bmatrix} -1 & -5 & -2 & 1 \\ 0 & 1 & 4 & -12 \\ 2 & 3 & 1 & -13 \end{bmatrix}$, the row reduced echelon form of A is	$\begin{bmatrix} 1 & 0 & 0 & -9 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -\frac{95}{25} \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & -9 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 9 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 9 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix}$	

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The determinant of $A = \begin{bmatrix} 2 & 3 & 17 & 2 \\ 0 & 0 & -2 & 1 \\ 0 & 2 & 1 & 4 \\ 0 & 0 & 0 & 3 \end{bmatrix}$ is	-12	6	24	None of these	
Which of the following matrices is not in row-echelon form	$\begin{bmatrix} 1 & 0 & 2 & 4 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 2 & 4 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 2 & 4 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 2 & 4 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	
If A is a square matrix and $A^2 = I$, then A =	A = I	A = -I	$A \neq I$	None of these	
Which of the following is not an elementary transformation?	Adding two rows	Squaring all the elements of the matrix	Adding two columns	Multiplying a row by a non-zero number	
Rank of a matrix $A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 0 \\ 1 & 0 & 0 & 0 \\ 4 & 0 & 3 & 0 \end{bmatrix}$ is	0	1	2	3	
A set of linear equations is represented by the matrix equation Ax = b. The necessary condition for the existence of a solution for this system is	A must be invertible	b must be linearly independent on the columns of A	b must be linearly dependent of the columns of A	None of these	
5. Consider the following two statements: i. The maximum number of linearly independent column vectors of a matrix A is called the rank of A. ii. If A is an n x n square matrix, it will be non-singular is rank A = n. With reference to the above statements, which of the following applies?	Both the statements are false	Both the statements are true	I is true but II is false	I is false but II is true	
6. In the matrix equation Px = q. which of the following is a necessary condition for the existence of at least one solution for the unknown vector x?	Augmented matrix [Pq] must have the same rank as matrix P	Vector q must have only non-zero elements	Matrix P must be singular	None of these	
67. The rank of the identity matrix of order n is:	n-1	N	N+1	None of these	
68. If A is a non-singular matrix of order n, then the rank of A is:	N	0	n-1	None of these	
69. If $A = \begin{bmatrix} 2 & 5 \\ 5 & 2 \end{bmatrix}$, then rank of A is	1	2	0	3	
70. The rank of matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ is	1	2	3	0	